A HOUSE DIVIDED: GETTYSBURG

One battle, finished. The A to Z plan from today to launch.

A product plan for taking the three days of Gettysburg, July 1-3, 1863, to a fully polished, shippable single-battle game: real-time battles of tens of thousands of simulated soldiers, command friction, smoke, sound and animation, built on the deterministic core that already exists and passes cross-platform tests.

Version 1.0 for approval · 2 November 2026 · Working title only (run a trademark search before any public use; the series name used by the reference game must never appear). Not legal advice. All dates, counts and thresholds are planning estimates or hypotheses unless a source is cited.

Contents

1. The plan on one page 2

2. Product definition: what players get 3

3. The quality bar: what "fully polished" means, measurably 4

4. The history we must get right 5

5. Technology: what exists, what remains, what is new for one big battle 7

6. Content production: art, terrain, units, effects 9

7. Animation and audio 10

8. Game design in detail 11

9. Handling the history with respect 12

10. Production plan: phases, gates, people 13

11. Quality control: the gate stack and Gettysburg-specific tests 15

12. Usage efficiency: how the build stays inside your budget 17

13. Clean room, rights and compliance 18

14. Launch, live service and the road to the full war 19

15. Risks and the descope ladder 20

Appendix A. The A to Z checklist 21

Appendix B. Scenario list (working IDs) 22

Appendix C. Open questions and honest limits 23

1. The plan on one page

The goal. Ship one battle, Gettysburg, as a complete, polished product rather than a thin slice of a big game. Every system a full Civil War game needs (simulation, command, AI, rendering, animation, audio, interface, tutorial, accessibility, localisation, store presence, post-launch support) is built once, at launch quality, for a single historical setting. That makes the battle both a product and the proof that the full-war game is buildable.

Why Gettysburg. It is the best-documented battle of the war, it has three distinct days that teach different lessons (meeting engagement and retreat; an attack on a defended ridge; a massed assault and a cavalry fight), its ground is protected and surveyed, and it is the single event players most want to command. Its documentation also means the historical-accuracy bar can be set high and tested.

What "done" means in one table

Dimension	Launch target (hypotheses, locked at gate M1 and again at the vertical slice)
Content	The three-day battle playable as a historical campaign of linked scenarios, plus about 12 stand-alone scenarios, a what-if set, a sandbox and a tutorial; an order of battle audited to regiment and battery level; real terrain for the whole battlefield.
Simulation	Per-projectile fire near contact, bundled and regiment-level fire elsewhere, one deterministic result shared by play, quick-resolve and replays; historical-replay mode that lands inside the documented casualty bands.
Fidelity	At least 20,000 individually animated soldiers visible at 60 fps on the recommended-spec reference PC, with smoke that blocks sight in the simulation.
Experience	Orders arrive late through couriers and signal flags; the player commands corps and divisions, not every man; the fight feels like the accounts.
Quality	Zero open P0 or P1 defects, crash-free beta sessions at or above 99.5%, accessibility audit passed, text proofread and historian-approved, ratings and store compliance done.
Respect	Advisory-board sign-off at each phase; the cause, the Black civilians of the town and region, and the cost are handled honestly (Section 9).

Timeline and cost shape

- **About 90 weeks to release candidate and about 100 weeks to launch** (roughly 22 months), with a vertical slice at week 26 and an alpha with all three days playable at week 52.
- That is a planning estimate. Part XII of the research volume benchmarks a conventional team of 40 to 80 people at 36 to 48 months for the whole war at high fidelity. One battle needs far less content but nearly the same technology depth, so this plan assumes agents do the engineering, tools, data and test work, and a small human and vendor lane does what agents cannot: art, motion capture, music, voice, historians, counsel, playtesting.
- A lean human core of about 8 to 14 people at peak plus contractors (Section 10), with explicit gates that stop spending if the technology spike or the slice fails.

Decisions needed from you (everything has a default)

#	Decision	Default
D1	Weekly usage budget and profile (Lean, Standard, Sprint)	Standard, re-tuned after 10 tickets
D2	Platform and engine host	Windows PC on Steam first; engine decided at gate M1 from a measured prototype
D3	Budget envelope for the human and vendor lane	Phase-gated; nothing committed before G0 and M1
D4	GPU workstation for renderer work and visual QC	Yours or a rented GPU runner from week 4
D5	Product name	Keep the working title; trademark search in P0
D6	Ambition on multiplayer	Single-player at launch; deterministic lockstep battles as a post-launch update
D7	Where the code lives	This repository's `a-house-divided/` directory now; move to its own repository before release

2. Product definition: what players get

2.1 Modes

Mode	What it is	Scale of work
Historical campaign	The three days as eleven linked scenarios with an overnight operational phase between days (move corps, set lines, bring up ammunition and reserves, choose where to hold). Play either army. Forces and arrival times follow the record; the player's choices can change what happens.	Core
Stand-alone scenarios	About 12 well-known actions, each playable alone in 20 to 60 minutes (see Appendix B).	Core
What-if set	Four to six pivots the record supports as plausible (for example a different arrival time for a cavalry division or a corps), each marked as a hypothetical.	Core
Sandbox	Pick a sector, forces, weather and time; place units; run it.	Core
Tutorial and drill ground	Teaches orders, formations, fire, morale, command delay and the map, using period drill language.	Core
Battlefield guide	A browsable, historian-approved guide to the ground, units and people, tied to the scenarios; a place to read about each regiment you command.	Core
Replay and photo mode	Deterministic replays, free camera, screenshot tools.	Core, near free from the architecture
Multiplayer	Lockstep battles are architecturally possible (results are deterministic) but deferred to a post-launch update.	Deferred

2.2 The player's role

The player is an army or corps commander working through friction. Orders go to corps and division commanders by courier or signal station, take time, and can arrive late or be misread. Commanders carry documented traits (aggressive, cautious, independent) drawn from the record, so the same order is carried out differently by different generals. Information about the enemy is delayed and partial. The player sees the field the way a headquarters would, with an option to turn on full information for those who want it.

2.3 Scale of the battle

- Engaged strength in the research volume (a figure it flags as contested between sources): roughly 94,000 Union and 72,000 Confederate. Not all are on the field at once; arrivals are scheduled by hour in the order of battle.
- Peak simultaneous action in the most intense scenarios is expected in the tens of thousands (the assault on Cemetery Ridge on the third day is documented at about 12,500 attackers in nine brigades, supported by 150 to 170 guns in the preceding bombardment; figures to be re-audited).
- The simulation uses the project's three fidelity classes: exact per-projectile fire near contact, bundled volleys at range, regiment-level rules for distant action. The class is a property of the scenario, not of the player's hardware, so results never depend on the machine.

2.4 Platforms and specifications (hypotheses)

Tier	Hardware (to be set by the prototype)	Target
Minimum	6-core CPU, 16 GB RAM, GPU around the class of an RTX 2060 or RX 5700 with 6 to 8 GB	1080p, 30 fps, reduced soldier counts
Recommended	8-core CPU, 32 GB RAM, RTX 3070 / RX 6800 class	1440p, 60 fps, full soldier counts with upscaling
High	Modern 8 to 16-core, 32 to 64 GB RAM, current high-end GPU	4K upscaled, highest counts and smoke resolution

Windows on Steam first. macOS and Linux and console ports are separate decisions after launch. Controllers are not supported at launch; the design is mouse and keyboard.

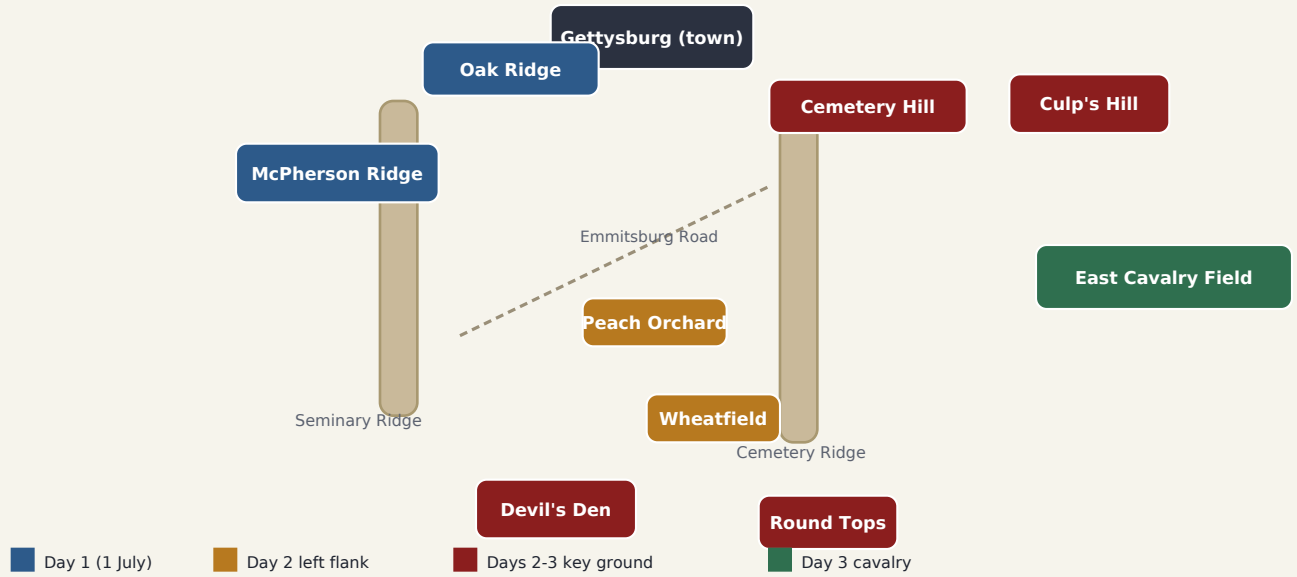
3. The quality bar: what "fully polished" means, measurably

"Polished" is a set of numbers and checklists, each tested before the gate that needs it. Values are hypotheses until the prototype measures what the hardware allows.

Area	Acceptance criterion	Verified by
Stability	Crash-free sessions at or above 99.5% across the beta; zero P0 and P1 defects open at release candidate; saves load across every patch; no desync between a replay and its recording.	Telemetry, nightly soak, golden replays on three operating systems
Performance	Recommended tier holds 60 fps with 20,000 visible soldiers in the heaviest scene (full smoke, rain, night skirmish and day assault); minimum tier holds 30 fps with reduced counts; simulation tick stays under 80% of its budget on the reference CPU.	Scripted camera flights on the GPU runner; tick benchmarks
Visual quality	Lighting, weather and smoke review signed off by the art lead and two outside reviewers at each phase; no placeholder asset visible at release; soldier variation measurable (at least six independent appearance axes).	Golden screenshot suite; art review checklist
Audio	Distance delay equals range over the speed of sound in every tested shot; aggregated fire does not exceed the voice budget in the heaviest scene; loudness within a stated range across modes; no clipping; mix reviewed on headphones and speakers.	Audio unit tests, capture diff tests, mix review
Animation	No foot-slide above threshold on slopes; every reaction in the coverage list present (flinch, take cover, aid wounded, fall, load, fire, march, wheel); no T-poses or pops in any scenario.	Automated pose checks; human review each cycle
History	Orders of battle audited at regiment and battery level against at least two independent sources per unit; every number in the game carries source and confidence; historical-replay mode lands inside casualty bands for the three days and for named regiments; historian sign-off.	Data validator; replay-band tests; advisory review
AI	No blobbing, chokepoint stalls, terrain blindness or silly retreats in the scripted failure suite; commanders behave in line with documented traits; each difficulty level is measurably different.	Nightly AI runs; scripted scenarios; playtest
Usability	At least 80% of new testers complete the tutorial unaided; no more than two blocking usability findings per playtest round; all text readable at the smallest supported size.	Moderated playtests; analytics
Accessibility	Remappable controls, scalable UI, colour-blind-safe palettes with shape cues, subtitles for all speech, reduced-motion and reduced-flash options, pause in every mode, adjustable game speed, difficulty options; audit against a recognised checklist passes.	Accessibility audit; user testing
Localisation	English plus a defined set of languages (set in D-decisions) fully translated, with in-context review and no truncation.	Localisation QA pass
Compliance	Age rating obtained; store requirements met including AI-content disclosure; data and privacy handled; asset provenance complete.	Compliance checklist; counsel
Respect	Advisory-board sign-off at each phase; sensitivity checklist passed; no gratuitous depiction.	Board review

4. The history we must get right

Schematic of the action zones (not to scale, north at top). Real terrain comes from survey data, not from this sketch.



4.1 The three days as game scenarios

Day	Actions that become scenarios (IDs in Appendix B)	Key tests of the simulation
1 July	Cavalry delaying action west of town; fighting on McPherson Ridge; the collapse north of town and the retreat through Gettysburg to Cemetery Hill; the first night's choices.	Meeting engagement with arrivals by hour; fighting retreat; command friction; the first big morale cascades
2 July	The Union left (Devil's Den, the Round Tops, the Wheatfield, the Peach Orchard, the advanced line along the Emmitsburg Road); Cemetery Ridge defence; the evening assaults on Culp's Hill and Cemetery Hill.	Terrain and cover (woods, rocks, walls); reinforcement timing; flank exposure; artillery on ridges; night fighting
3 July	Morning fighting on Culp's Hill; the great bombardment; the massed assault on the centre of Cemetery Ridge; the cavalry fights east of the field and on the Union left.	Counter-battery and smoke; long-range fire against advancing lines; canister at close range; massed assault versus fortified ridge; cavalry

4.2 What the data layer must hold for this battle

- **Order of battle by hour:** corps, divisions, brigades, regiments, batteries; arrival and departure times; strengths; commanders; weapons mix (smoothbore or rifled, caliber, breech-loaders); training and experience levels. The research volume's Gettysburg record is a single, partly verified record built from one headline source and conflicting tallies, so this audit is new work.
- **Terrain:** real elevation (public-domain lidar-based elevation data where coverage allows), historical land cover for July 1863 from period maps and surveys, roads, fences and walls, woods, orchards, fields, creeks, rock outcrops, farm buildings and the town's streets.
- **Timeline events:** fixed historical triggers (arrivals, famous orders, wounding of commanders) with times and their evidence tags, so the simulation can be validated against the record.
- **Casualty targets:** by day, by corps and for named regiments, with the uncertainty the sources actually have.
- **People:** commanders with documented traits and relationships, each backed by an act in the record.
- **Weather and conditions:** heat, humidity, wind and visibility for each day, from period reports.

4.3 Known data conflicts to resolve before they enter the game

Both audits of the research volume found conflicts that touch this battle. Each becomes a regression fixture: the validator fails the build if an unresolved value is ever held as verified.

Item	Conflict in the volume	Action
Engaged strengths and casualty totals	Headline source figures differ from older tallies (for example Confederate casualties of about 28,000 in one source and about 23,000 in older ones)	Resolve per unit from the regimental record; store ranges
Hood's step-off, day 2	About 4:00 p.m. and about 4:30 p.m. in the same pages	Resolve from official reports and later studies; record both until settled
Day 3 assault step-off	Ledger says about 2 p.m.; the text implies about 3 p.m.	Resolve; choose the evidenced time
Stonewall Brigade, Florida and Georgia brigades at Gettysburg	Brigades and commanders conflated in the commander roster	Rebuild from an audited order of battle
Union reserve artillery and gun counts at the bombardment	Range of 150 to 170 guns; other figures differ	Audit by battery
Day-by-day casualty split	Components do not always sum to totals	Recompute from unit data; validator sum rule

4.4 Source plan (candidate sources, to be confirmed as available and consulted)

The research volume was compiled mostly from web encyclopedias and a battlefield-trust site, and it states its own limits. Gettysburg deserves primary and specialist sources. Candidates: the Official Records of the war (public domain); the park service's historic resource and orders-of-battle materials; the Bachelder papers and maps; the Warren survey map; specialist regimental-strength and loss compilations; the established multi-volume studies of the three days and the campaign; the Library of Congress photographs. None has been consulted yet; the audit's job is to confirm each source exists in the form assumed, record its edition and licence, and cite page-level evidence in the data.

5. Technology: what exists, what remains, what is new for one big battle

5.1 Already built and proven (this repository, passing continuous integration)

- **core-math:** fixed-point numbers, integer-only trigonometry tables, vectors, named random streams.
- **sim-state:** deterministic agent store, spatial grid, canonical state hash, replay format.
- **hd-sim:** a headless harness; three golden replays verified on Linux, Windows and macOS in continuous integration. Baseline: 20,000 simulated agents in the toy scenario average about 11.8 ms per tick (99th percentile about 17 ms) on a 4-core cloud machine, against a 50 ms tick at 20 Hz.
- **Data layer:** schemas, validator with provenance rules, 5 battles and 30 commanders (none yet marked verified), 13 known-error regression fixtures.
- **Process:** forbidden-terms scan, no-process-text lint, path-filtered CI, ADRs, specs SPEC-001 to SPEC-005, and the ticket list T-014 to T-024.

One piece of work is open: the independent review's hardening of the three crates (a verify-time hang on crafted replays, looser replay validation, and tests for 15 mutations the original tests miss) is saved locally and not yet finished. It finishes first (Section 14).

5.2 To build for this product

Layer	Work	Status / spec
Battle rules	Movement, formations and drill; weapons and ballistics; hit tests, cover, casualties; morale; command messages; ammunition; fatigue and heat	SPEC-004, SPEC-005, tickets T-014 to T-024
Gettysburg scale	Fidelity-class scheduler (exact near contact, bundled at range, regiment level elsewhere); sector streaming of terrain and agents; arrivals by hour; scenario loader for the order of battle	New specs in P1
Command layer	Courier and signal messages with latency and loss; subordinate interpretation by trait; information model with delayed, partial sightings	New specs in P2
Operational layer	The overnight phase: lines, reserves, ammunition and ambulance trains, orders for next day	New specs in P3
Smoke and weather	Smoke density grid (simulation) plus rendering; wind; rain; heat effects on fatigue; time of day	New specs in P2
AI	Tactical AI, operational AI, commander personalities from the record; failure suite	New specs in P2 to P3
Historical replay mode	Feed documented orders and arrival times into the simulation and compare outcomes with the record	New spec in P2; the core QC tool for history
Rendering	Tiered crowd renderer, terrain at battlefield scale, volumetric smoke, lighting, vegetation, camera	Prototype at M1; engine host decided there
Audio	Acoustic propagation with distance delay and terrain shadowing; aggregated musketry and artillery; mix	Prototype at M1
Animation	Motion-matched near soldiers, baked clips with phase offsets for the mass, matched interactions, terrain adaptation	Capture plan in Section 7
Interface	Command views, orders, regimental cards, timeline, map overlays, replay and photo tools	Design in P2; build in P3 to P4
Tools	Scenario editor, data compiler, balance and calibration harness, telemetry	Grows with each phase

5.3 Engine decision (gate M1)

The research volume's technology analysis recommends a commercial engine as host with a self-owned simulation core and a custom crowd-render module, with a custom renderer as the fallback. That remains a recommendation until the week-10 prototype measures 20,000 animated soldiers in three tiers, smoke and a terrain slab on real hardware. The simulation core is engine-independent by design, so the choice affects rendering, tooling and hiring, not the rules or the tests.

Why this prototype first. The two things most likely to sink a battle game are simulation scale with deterministic results and crowd rendering at target fidelity. The first is already measured and green. The second is measured at gate M1. If M1 fails, the plan cuts visible soldier counts or switches the rendering approach before any art is bought.

6. Content production: art, terrain, units, effects

Counts below are planning estimates to size vendor quotes. They shrink or grow once the audited order of battle exists.

Asset class	Estimated scope	How it is made	Lane
Battlefield terrain	About 100 square kilometres of surveyed ground at several detail levels, including the town and approach roads, plus the east cavalry ground; hand-polish of about a dozen key sectors (for example the Round Tops, the Wheatfield, the ridge in front of Cemetery Ridge, Culp's Hill)	Survey elevation, period maps, aerial analysis; procedural vegetation from historical land cover; artist polish	Mixed
Structures	About 150 farm buildings, barns, fences, walls and the town's buildings at three levels of detail, with a few dozen hero buildings	Kit-based modular pieces; photogrammetry where permitted	Vendor
Soldier kit	Modular uniform pieces (headgear, coats, trousers, equipment) with texture variation, enough for the documented range: regulation blue, mixed early-war dress, Confederate dress ranging from regulation grey to home-made, plus distinctive units documented in the record	Reference from period photographs, regulations, museum collections; artist-built kit	Vendor + lead artist
Weapons	Rifled and smoothbore muskets, carbines, sidearms, edged weapons; field guns of the documented types; limbers, caissons, wagons, ambulances	Reference from museum pieces; artist-built	Vendor
Animals	Cavalry horses, artillery teams, draught animals	Artist-built; mocap with trained horses where practical	Vendor
Flags and signs	National, state and Confederate flags, regimental colours, headquarters flags, signal flags, unit markers	Reference from museum collections; many are documented	Artist
Effects	Muzzle flash, smoke (simulation-driven), shell bursts, canister, ricochets, dust, rain, fire in buildings, blood in restrained form	Tech art	Lead tech artist
Interface art	Period-inspired map style, regimental cards, icons, fonts	Designer + historians for period authenticity	Designer
Audio assets	Field recordings, foley, voices, music (Section 7)	Section 7	Vendor

Visual direction

Sober, painterly realism based on period photographs and the landscape as it was in July 1863, which differs from the park today (the park service restores some of the historical land cover; the plan uses period evidence, not the modern view). Soldiers differ visibly from each other (age, build, dress wear, equipment) and from regiment to regiment; the same man is not cloned across the field.

7. Animation and audio

7.1 Animation

- **Capture:** sessions with reenactors performing period drill: loading and firing in the manual's counts, marching, wheeling, advancing under fire, taking cover, falling, aiding wounded comrades, officers rallying men, artillery crew drill, cavalry riding and dismounting. A defined capture list (about 300 clips) with named performers and consent agreements.
- **Near the camera:** motion matching over the captured library for fluid transitions; context-aware reactions (flinch at close bursts, hesitate before an advance, look to officers).
- **The mass:** baked clip libraries with randomised phase and variation so formations do not move in lockstep.
- **Interactions:** a small set of paired animations (carrying wounded, officers rallying, gun crews working a piece) selected by the combat system and tied to its outcomes.
- **Terrain:** foot placement on slopes, in ditches and over fences and walls.
- **Restraint:** deaths and wounds are depicted honestly and soberly, not as spectacle; ratings guidance applies.
- **R&D, not a ship dependency:** learned physical character control.

7.2 Audio

- **Sound as simulation:** finite speed of sound (guns heard after the smoke is seen), terrain echo, wind, and acoustic shadow, the documented effect that the loudest battles could be inaudible to nearby commanders. The third day's bombardment is documented as audible at very great distances; the exact reports are to be verified before they are used in the game.
- **Aggregated musketry:** individual shots combine into rolling fire that changes with distance, direction and terrain.
- **Voices:** officers' commands from the drill manuals, sergeants' calls, cheers and the Confederate and Union battle cries, recorded by consenting performers under fair agreements.
- **Music:** period tunes from public-domain sources arranged and performed by contracted musicians; an original score by a human composer; adaptive layers driven by the tension of the battle.
- **Mix:** a dynamic mix with a fixed voice budget and a loudness standard; reviewed on headphones and speakers.
- **Interface and ambience:** a restrained, period-flavoured UI soundscape; field and town ambience at dawn and dusk.

8. Game design in detail

8.1 Battle

- Soldiers are individual agents in regiments; regiments in brigades, divisions and corps. Orders go through the chain of command with delay.
- Fire is the main killer; bayonet fights are rare and short, as in the accounts. Charge-and-collision is not the dominant way a battle ends.
- Morale is a regiment property built from casualties, officer losses, supply, fatigue, terrain, flanking and seeing neighbours break. The player sees symptoms (slowing fire, men looking back, ranks thinning) and a coarse indicator, not a hit-point bar.
- Fortification: troops dig in when they stop; stone walls and rail fences give cover; woods break formations; the ridge advantage is real.
- Artillery works through batteries with ammunition, limbers and crews, and is vulnerable to counter-battery fire and infantry assault; canister at close range is devastating.
- Cavalry screens, raids, charges and fights on foot.

8.2 The overnight phase

Between days the player sets lines, assigns reserves, sends ammunition and ambulance trains forward, and issues the next day's orders to commanders. Fatigue and ammunition carry over; units that fought hard need time. This phase is where the army-level decisions of the record (where to stand, whether to attack, which corps to bring up) become the player's choices.

8.3 AI

- Three layers: operational (corps and division movements), tactical (battle plans executed by regiments) and a personality layer from the record: an aggressive commander plays differently from a cautious one.
- The AI uses the same message interface as the player, sees only what its headquarters would, and does not cheat. Difficulty changes behaviour (initiative, risk tolerance), not hidden bonuses.
- Simulation lookahead (running quick forecasts of the deterministic core) evaluates plans. Any language-model feature, if added, sits behind a strict command schema and is never a ship dependency.

8.4 Tutorial, difficulty, scoring

- A drill-ground tutorial of about 40 minutes introducing orders, formations, fire, morale, courier delay and the map; contextual hints; a skippable tutorial for experienced players.
- Difficulty levels change AI initiative, information accuracy and the player's tolerance for delay, not enemy statistics. Historical accuracy options toggle what the player may do.
- Scoring is by objectives and by comparison with history (ground held, casualties against the record, units saved), not by kill counts alone.

8.5 Interface

- Command view: a period-map style with regimental markers, confidence shading for enemy information, an orders panel, a timeline, an overlay for terrain, cover and supply.
- Field view: free camera, unit cards with the regiment's real name and short history, minimal HUD.
- Every system has a text and accessibility mode; every screen is navigable by keyboard.

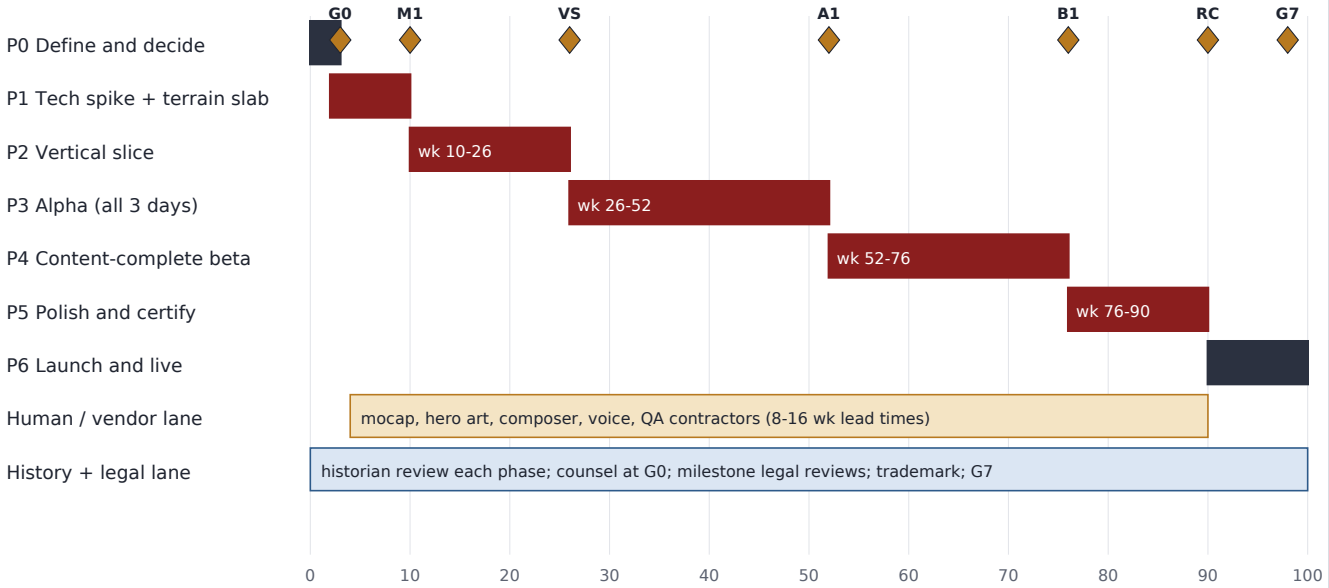
9. Handling the history with respect

Gettysburg is sacred ground for many people and a site of living memory. The product treats it that way. These commitments are product requirements, each with a review gate:

- **The cause.** The war was fought over slavery; the Confederate states' own declarations said so. The framing, briefing text and political context say so and do not romanticise the cause.
- **The people of the place.** The town's Black residents, and the free and enslaved Black people of the region seized or driven off during the Confederate invasion of Pennsylvania, are part of the story. The research on this topic is to be verified with specialist historians before it enters the game; it must not be left out.
- **The cost.** The record shows enormous casualties, and the sound and image of death are handled with restraint. A memorial and epilogue (the national cemetery and the November dedication) frames the battle.
- **The myths.** The popular narrative of the third day's assault and the postwar memory of the battle are contested; the guide and briefings present evidence, and where the record is disputed they say so.
- **Consultation.** A historical advisory board (historians, park interpreters, and voices from descendant communities) reviews content at the vertical slice, alpha, beta and release candidate with the right to demand changes. The plan assumes outreach to the park service and its partner foundation for advice; any use of their material or name requires written permission.
- **Symbols.** The Confederate battle flag appears as it did on the field and in the order of battle, with historical context and without glorification; platform and regional rules are checked at G7.

10. Production plan: phases, gates, people

Calendar view in weeks. Planning estimates with low confidence beyond the vertical slice; re-baselined at every gate.



Phase (weeks)	Agent work	Human / vendor / history lane	Exit gate
P0 Define (0-3)	Finish hardening and push; product definition; schemas for the full order of battle; scanners; ledger; engine spike brief; counsel brief	Engage counsel; engage a lead historian; draft charter; name check; decide D1-D7	**G0:** charter approved, decisions recorded, CI green
P1 Tech spike + terrain slab (2-10)	Battle-rules tickets T-014 to T-024 (movement, ballistics, morale v0); fidelity-class scheduler; 20,000-agent crowd render test; smoke and acoustics prototypes; one real-terrain slab (about 4 square kilometres around the Round Tops and the Wheatfield)	GPU machine ready; first reenactor reference session scoped; art-direction frames from the lead artist	**M1 (wk 10):** 20,000 agents in budget on target hardware; render prototype measured; engine decided; thresholds locked
P2 Vertical slice (10-26)	The assault on Cemetery Ridge, the most iconic hour, built to near-final quality: command friction, couriers, smoke, bombardment, morale, tactical AI, quick-resolve, historical-replay mode, first animation and audio at final pipeline, UI skeleton, calibration against the record	First motion-capture session; first composer demo; historian review of the slice; milestone legal review (G6)	**VS (wk 26):** slice meets the quality bar for its scope; calibration within bands; advisory sign-off
P3 Alpha (26-52)	All three days playable with grey-box or early art; terrain for the whole field; audited order of battle loaded; overnight phase; operational and tactical AI; scenario editor; tutorial draft; save and load; telemetry	Environment and kit art production; second mocap session; voice casting; localisation plan; playtests start	**A1 (wk 52):** every scenario plays end to end by bot and by you; nightly suite stable for 14 days; G6
P4 Beta (52-76)	Content complete: art, animation, audio, UI, text frozen; balance by simulation; performance tiers; accessibility build; localisation delivery; AI tuning	Art and audio delivery; closed beta; advisory review; rating submission; store page and trailer	**B1 (wk 76):** content complete, no placeholder visible, crash-free target in beta
P5 Polish + certify (76-90)	Bug and performance hardening; golden-screenshot and audio regression; final balance; installer and updater; release candidate builds	Compliance, ratings, store review; marketing demo; press kit	**RC (wk 90):** zero P0/P1; quality bar met
P6 Launch + live (90-100)	Launch build; patch pipeline; telemetry dashboards; hotfix rota	G7 clearance, launch, community support	**G7 (wk 98):** counsel and producer clearance

10.1 Team and effort (planning estimates, to be replaced by vendor quotes)

Role	Peak size	Notes
Director and product owner (you)	1	Decisions at gates; about 20 minutes a week plus gate reviews and playtests
Technical direction and engineering	Agent sessions + 1 human technical reviewer from P1	Agents do the engineering; the human reviewer owns architecture decisions, security review and the engine host integration
Art	1 lead + 4 to 8 contract artists (environment, characters, tech art, VFX)	Largest cost line
Animation and capture	1 animator + capture studio sessions + reenactor performers	Capture sessions scheduled in P2 and P3
Audio	1 audio designer + composer + voice performers + field recordist	Contracted
History	1 lead historian + 3 to 5 advisors + battlefield specialists	Review at each gate
Design and UI	1 designer + 1 UI/UX contractor	Agents implement; designers decide
QA	2 to 4 contract testers from P3, a larger beta cohort	Plus the automated stack
Production, legal, community	1 producer, counsel (hourly), 1 community manager from P4	Producer owns the schedule and vendor contracts

Effort in person-months for the human lane at peak is of the order of 150 to 250 person-months across the project (planning estimate, wide error bars); the agent lane replaces what a conventional team would spend on engineering, tooling, data entry and test automation.

11. Quality control: the gate stack and Gettysburg-specific tests

Seven QC layers: fast and cheap run most often; slow and human run least often



Layer	When	Checks	Pass criterion
Q0 local	Before every commit	Formatting, strict lints, tests for touched crates, forbidden-terms scan, no-process-text lint	All pass in under a minute
Q1 pull request	Every PR	Unit and integration tests, data validation, licence and similarity scans, determinism smoke, benchmarks against baseline, coverage floor on rules crates	All pass; no benchmark regression above 10%
Q2 merge	Every merge to main	Three operating systems, two compilers: replay hashes identical; property tests; fuzzing on parsers and loaders	Hashes identical across the matrix
Q3 nightly	Unattended	1,000 headless AI battles on rotating scenarios with invariants; historical-replay bands; performance scenes; soak; memory and leak checks	No violation; bands met
Q4 cycle	Every two weeks and every gate	Mutation testing on rules crates; independent review of the cycle's diff; clean-room red-team pass; bot playthroughs; 5% data re-verification; golden-screenshot and audio diffs	Mutation score at or above 80%; no open red-team finding
Q5 human	Every cycle	Your playtest; moderated external playtests; accessibility and sensitivity review; advisory board at VS, A1, B1, RC	Sign-off recorded
Q6 release	Before launch	G7: trademark, patent, compliance, ratings, store requirements; installer and update tests; clean-machine installs	Counsel and producer clearance

Gettysburg-specific tests

- **Historical-replay bands.** Replay the documented orders and arrivals for each scenario; the simulation's casualties by day, by corps and for named regiments must fall inside the sourced bands. Examples to test: the heavy losses of the Iron Brigade and the Minnesota regiment on days 1 and 2, the third day's assault losses, the east cavalry field. Bands carry the confidence of their source, and unverified bands report without gating.
- **Order-of-battle audit.** A script checks that every unit has two independent sources, valid dates, no commander in two places at once and strengths that sum.
- **Terrain truth.** Sight-line and cover tests at documented vantage points (for example visibility between named positions on the ridges), line-of-fire tests along the third day's approach.
- **Timeline tests.** Fixed triggers (arrivals, commander wounds) happen at their evidenced times when the historical orders are replayed.
- **Scenario completion bots.** Bots finish every scenario in every mode; failures are logged with replays.
- **Golden screenshots and audio captures** for fixed camera positions and audio events, diffed each cycle.
- **Compatibility matrix.** Driver and hardware matrix for the recommended and minimum tiers; clean-install and upgrade tests; save compatibility across patches.
- **Localisation QA.** String freeze, pseudo-localisation tests, in-context review, truncation checks.

Defect policy

- P0 (crash, desync, data corruption, contamination) blocks every merge. P1 is fixed within the cycle. P2 is scheduled; nothing in P0 or P1 ships.
- Every escaped defect gets a regression test and a one-line cause in the ADR log before it closes. Tests are never weakened, skipped or quarantined to turn a gate green.
- Authors of tests and code differ; reviewers are a different agent (and a different model family where possible). Thresholds change only by ADR and your approval at a gate.

12. Usage efficiency: how the build stays inside your budget

1. **Extract once.** Research becomes validated data and is never re-read as a PDF by a build session. (The 524-page research volume is about 1.7 million characters.)
2. **One ticket, one short session.** Context is the ticket, its spec section and a repository map of about 150 lines. Hand-offs are short notes, not transcripts.
3. **Cheapest competent model first.** Mechanical work on small models, escalating on failure; strong models for the core, specs, review and red-teaming.
4. **Heavy checks in CI or overnight, not in chat.** Local checks print failures only.
5. **Stop-loss.** A session at twice its estimate stops and re-plans. No polling; CI results wake the session.
6. **Tests from specs, by a different agent than the implementer.** It is cheaper than debugging vague behaviour.
7. **Budget profiles.** Lean (one core session plus one mechanical), Standard (one core plus up to two in parallel), Sprint (up to four, only before a gate).
8. **Usage ledger.** One row per accepted ticket (estimate, actual, defects, rework); a weekly summary; a rising trend triggers a process fix before more feature work.
9. **Your time.** About 20 minutes a week for the ticket queue and decisions; five or six gate reviews; playtests; everything else has a default.

Planning allocation of a week's usage (re-set after ten tickets of real data): implementation 40%, review and QC 20%, data and research 15%, specs 10%, reserve 15%.

13. Clean room, rights and compliance

> **Not legal advice.** Counsel approves the independence charter at G0 and reviews at each milestone. The plan's controls make independence demonstrable; they do not guarantee a legal outcome.

- **Independence.** Original design, original data schemas, original names and original interface. Nothing from the reference game's code, data, models, textures, sounds, text or user interface is used. Documents describing that game stay in the reference environment (the repository's `audits/` folder) and reach specs only as sanitised abstract lessons through a human-reviewed gate. Implementation sessions start fresh and see only specs, datasets and permitted material.
- **Evidence trail.** Provenance manifests for every artefact; independent-creation dossiers per system; immutable logs; scans (forbidden terms, similarity, licences) on every merge.
- **Name and branding.** The series name used by the reference game must not appear in the product or its marketing; trademark and patent searches start in P0, not at the end.
- **Historical imagery and maps.** Federal works and period maps are generally usable, but each item is rights-checked; monuments and some museum photographs carry separate rights. Park service materials and names need written permission.
- **Music and voice.** Public-domain tunes arranged anew; composer and performers under written agreements covering digital replicas; no use of generated voices of real people.
- **AI-generated material.** Meaningful human authorship for key creative assets (characters, signature art, the score, the sound identity); generators used only under reviewed terms; store disclosure rules checked at release.
- **Ratings and platforms.** Age ratings for historical violence; store requirements; privacy and data handling; regional compliance.
- **Contamination response.** Stop and quarantine, preserve evidence, assess with counsel, rebuild from the clean spec with fresh sessions, fix the process, never edit logs.

14. Launch, live service and the road to the full war

14.1 First 90 days from approval

Window	Work	Exit
Days 1-7	Finish the hardening work in the stash (spec amendments are already pushed); decisions D1-D7; counsel and lead historian engaged; GPU machine arranged; trademark check; CLAUDE.md, ADRs and ledger updated	CI green; decisions recorded
Days 8-21	Implement T-014 to T-017 (segment query, rules crate and loader, locomotion and terrain, drill and slots); extraction pass 2 and second-source verification of day-1 and day-2 records; art-direction frames	Movement scenarios deterministic and in the golden suite
Days 22-45	T-018 to T-021 (state and drill, steering, ammo and damage tables, projectile flight); real-terrain slab pipeline; crowd render prototype on the GPU machine; acoustics and smoke prototypes	Flight tests against independent references; slab renders
Days 46-70	T-022 to T-024 (fire phase, hit test and casualties, calibration harness); fidelity-class scheduler; fixed-trigger replay mode skeleton	Calibration harness reports against the first bands
Days 71-90	Gate M1 review: measurements, engine decision, locked thresholds, re-baselined plan; vertical-slice design (the assault on Cemetery Ridge) written and approved	**M1 passed or the plan is cut or re-aimed.**

14.2 Launch and after

- **Marketing.** A public demo of the vertical slice scenario in the beta phase; a store page and wishlist campaign from the alpha; a trailer from real gameplay; a press kit; a community channel; outreach to history and wargaming communities.
- **Pricing and release.** To be decided by you at B1 with the market in view; the plan does not assume a price.
- **Live support.** A hotfix rota for the first two weeks; weekly patches for a month; telemetry dashboards for crash rate, performance and tutorial completion.
- **Post-launch content.** Lockstep multiplayer battles; the remaining cavalry and smaller actions; more what-if scenarios; modding tools (data formats are already open and documented).
- **The path to the full game.** Each further battle is a data-and-content job on the same engine; campaign layer work (supply, politics, the map) from the earlier plan resumes afterwards. Gettysburg's success, not a promise, decides whether and how that proceeds.

15. Risks and the descope ladder

Risk	Trigger	Mitigation
Crowd rendering misses target at 20,000 soldiers	M1 measurement	Reduce counts, change approach, or switch host before buying art; renderer verdict may extend to week 14
Deterministic simulation too slow at battlefield scale	Tick above budget	Fidelity classes; tiered update; data-oriented layout; reduce simultaneous exact-fire area
Cross-platform drift	Any replay hash mismatch	Fixed-point authority; strict lints; matrix on every merge; P0
Historical data errors reach the game	Sampled re-verification above 5% error	Two-source rule; exclude unverified; re-do the batch; historian audit
Vendor lead times slip	Capture or art late	Placeholder-first pipeline; early vendor contracts; milestone payments
Art and animation cost exceeds budget	Quote review at P2	Reduce kit variety, shrink the hero-asset list, reuse across regiments; cut capture list
Sensitivity failure	Advisory objection	Early framing decisions; reviews at each gate; stop-ship authority
Contamination finding	Any scan hit	Quarantine, rebuild from the clean spec with fresh sessions
Usage overrun	Ticket above 2x estimate	Stop-loss; Lean profile; descope ladder
Scope creep	Unplanned feature requests	Mode list frozen at G0; changes need an ADR

Descope ladder (cut in this order; never the gates)

1. What-if scenarios beyond the first two.
2. The overnight phase's secondary logistics (keep lines and orders).
3. The tutorial's optional modules (keep the core one).
4. Capture list breadth (hand-keyed secondary actions).
5. Far-tier visual fidelity and some weather states.
6. Localisation to fewer languages at launch.
7. Stand-alone scenario count.

> **Never cut:** determinism, the QC gates, the clean-room gates, historical-replay calibration, data provenance rules, accessibility basics, and the respectful handling of the subject.

Appendix A. The A to Z checklist

	Item	What done looks like	Phase
A	Accessibility	Audit passed; remapping, scaling, subtitles, colour-blind palettes, reduced motion	P3-P5
B	Battlefield terrain	Real terrain for the whole field, polished key sectors, sight-line tests	P1-P4
C	Command friction	Couriers, signals, delay, interpretation by trait, partial information	P2-P3
D	Data and history	Audited order of battle at regiment and battery level; provenance everywhere	P0-P4
E	Engine host	Decided at M1; integration tested on target hardware	P1
F	Fidelity classes	Exact, bundled and regiment-level fire with calibration between them	P1-P2
G	Gates	G0 to G7 passed and documented	All
H	History review	Advisory board sign-off at each phase	All
I	Interface	Command and field views, map overlays, regimental cards, photo mode	P2-P4
J	Journals and briefings	Historian-reviewed briefings, epilogue, guide text	P3-P4
K	Kit	Uniforms, equipment, weapons, flags, artillery pieces and wagons	P2-P4
L	Localisation	Delivered languages with in-context review	P4-P5
M	Music	Original score, arranged period tunes, adaptive layers	P2-P4
N	Naming	Trademark and patent clearance; no reference-game names anywhere	P0, G7
O	Overnight phase	Operational planning between days	P3
P	Performance	Tiers met; benchmarks in CI; regression budgets	All
Q	QA	Full gate stack, playtests, compatibility matrix, soak	All
R	Release	Installer, updater, store page, ratings, launch build	P5-P6
S	Sound	Acoustic model, aggregated fire, voices, mix to loudness standard	P1-P4
T	Tutorial	Drill-ground tutorial; at least 80% unaided completion	P3-P4
U	Usage ledger	Every ticket recorded; weekly summary; budget respected	All
V	VFX and smoke	Simulation-driven smoke and volumetric rendering; effects polish	P1-P4
W	Weather	Rain, heat, wind, time of day affecting play	P2-P3
X	eXtensibility	Open data formats and documented scenario editor for modders	P3-post
Y	Year-one updates	Patch plan, lockstep multiplayer, extra scenarios	Post-launch
Z	Zero P0/P1 at launch	Release-candidate bar met; telemetry and hotfix rota ready	P5-P6

Appendix B. Scenario list (working IDs)

ID	Scenario	Day	Teaches
GB-01	Cavalry delaying action west of town	1	Delay, fighting retreat, arrivals by hour
GB-02	McPherson Ridge and the first infantry fight	1	Meeting engagement, flanking
GB-03	North of town: the collapse and retreat to the heights	1	Morale cascade, retreat, command friction
GB-04	Evening on Cemetery Hill (operational choices)	1	Defence of key ground; overnight phase
GB-05	The advanced line and the Peach Orchard	2	Exposed salient, artillery, crossfire
GB-06	Devil's Den and the Round Tops	2	Rough terrain, cover, hasty defence
GB-07	The Wheatfield	2	Local reserves, repeated assaults, fatigue
GB-08	Cemetery Ridge defence, evening	2	Reinforcement timing, desperate counterattack
GB-09	Culp's Hill (night and morning)	2-3	Entrenchment, night fighting, command friction
GB-10	The great bombardment	3	Counter-battery, smoke, ammunition, acoustics
GB-11	The assault on Cemetery Ridge	3	Fire against advancing lines, canister, morale
GB-12	East cavalry field and the Union left flank	3	Cavalry combat, screening
GB-W1..W4	What-ifs (marked as hypothetical)	varies	Command decisions and their consequences

Times, forces and objectives for each scenario come from the audited order of battle. The list is working: the audit may merge or split scenarios.

Appendix C. Open questions and honest limits

- **Not yet verified.** No outside source or primary document was consulted for this plan. The Gettysburg details above come from the research volume, which itself states and shows data conflicts; the data audit in P0 to P2 exists to correct them.
- **Estimates.** Dates, counts, person-months and thresholds are planning estimates; confidence is low beyond the vertical slice.
- **Scale of art and animation.** The largest cost and schedule uncertainty is human production quality, not engineering; vendor quotes in P0 to P2 replace the estimates here.
- **Rights.** Park service materials, museum images and any real people's likenesses need explicit permission.
- **Counsel.** The clean-room approach has no court endorsement for an agent-assisted process; counsel must decide whether the evidence trail is persuasive.
- **This plan's author.** The session that wrote it had read documents about the reference game, so under the project's own rules it belongs in the reference environment and cannot implement anything; implementation sessions must start fresh from the specs.